NAME

outb, outw, outl, outsb, outsw, outsl, inb, inw, inl, insb, insw, insl, outb_p, outw_p, outl_p, inb_p, inw_p, inl_p − port I/O

LIBRARY

Standard C library (*libc*, *−lc*)

SYNOPSIS

#include <sys/io.h>

unsigned char inb(unsigned short *port***);**
unsigned char inb_p(unsigned short *port***);**
unsigned short inw(unsigned short *port***);**
unsigned short inw_p(unsigned short *port***);**
unsigned int inl(unsigned short *port***);**
unsigned int inl_p(unsigned short *port***);**

void outb(unsigned char *value***, unsigned short** *port***);**
void outb_p(unsigned char *value***, unsigned short** *port***);**
void outw(unsigned short *value***, unsigned short** *port***);**
void outw_p(unsigned short *value***, unsigned short** *port***);**
void outl(unsigned int *value***, unsigned short** *port***);**
void outl_p(unsigned int *value***, unsigned short** *port***);**

void insb(unsigned short *port***, void** *addr***[.***count***],**
    unsigned long *count***);**
void insw(unsigned short *port***, void** *addr***[.***count***],**
    unsigned long *count***);**
void insl(unsigned short *port***, void** *addr***[.***count***],**
    unsigned long *count***);**
void outsb(unsigned short *port***, const void** *addr***[.***count***],**
    unsigned long *count***);**
void outsw(unsigned short *port***, const void** *addr***[.***count***],**
    unsigned long *count***);**
void outsl(unsigned short *port***, const void** *addr***[.***count***],**
    unsigned long *count***);**

DESCRIPTION

This family of functions is used to do low-level port input and output. The out* functions do port output, the in* functions do port input; the b-suffix functions are byte-width and the w-suffix functions word-width; the _p-suffix functions pause until the I/O completes.

They are primarily designed for internal kernel use, but can be used from user space.

You must compile with **−O** or **−O2** or similar. The functions are defined as inline macros, and will not be substituted in without optimization enabled, causing unresolved references at link time.

You use **ioperm**(2) or alternatively **iopl**(2) to tell the kernel to allow the user space application to access the I/O ports in question. Failure to do this will cause the application to receive a segmentation fault.

VERSIONS

outb() and friends are hardware-specific. The*value* ar gument is passed first and the *port* ar gument is passed second, which is the opposite order from most DOS implementations.

STANDARDS

None.

SEE ALSO

ioperm(2), **iopl**(2)